Synopsis

Project Title :

Online Purchase Behavior Classification using Machine Learning

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1. Problem Description:

- In the field of e-commerce, understanding customer behavior is essential for increasing sales and improving user experience. This project aims to predict whether a visitor will make a purchase based on their online browsing behavior. The dataset includes features such as number of pages visited, time spent on pages, bounce rates, and visitor type.
- The dataset will undergo data preprocessing, including handling missing values, removing duplicate records, and transforming categorical data into numerical form. The target variable 'Revenue' indicates whether the user completed a purchase (True/False).
- Machine learning algorithms like Logistic Regression and K-Nearest Neighbors (KNN) will be applied after preprocessing and feature scaling. The model will be evaluated using accuracy, precision, recall, and F1-score.

2. Steps of Implementation:

a) **Data Collection:**
Obtain dataset containing user behavior (clicks, browsing time, etc.)

b) **Data Preprocessing:**

   a. Handle missing values
   b. Normalize/scale data
   c. Encode categorical variables

c) **Feature Selection:**

   a. Remove duplicate or irrelevant features
   b. Select important behavioral attributes

d) **Model Building:**

   a. Apply Logistic Regression
   b. Apply K-Nearest Neighbors (KNN)

e) **Model Training:**

   a. Split dataset into training and testing sets
   b. Train models on training data

f) **Model Evaluation:**

   a. Compare performance of both models

g) **Prediction:**

   a. Predict whether a user will purchase or not

Fig. 1 Sample architecture

3.Expected Modules/ Libraries to be used :

- **NumPy:** Used for numerical operations and array handling
- **Pandas:** Data manipulation and preprocessing
- **Matplotlib / Seaborn:** Data visualization
- **Scikit-learn:** Machine learning models (Logistic Regression, KNN)
- **StandardScaler:** For data scaling
- **Train_test_split:** For splitting dataset

4.Expected Evaluation metrics and Plots to be used:

Metrics:

- **Accuracy:** Overall correctness of model
- **Precision:** Correct positive predictions
- **Recall:** Ability to find all positive cases
- **F1-Score:** Balance between precision and recall
- **Plots:**

a) Confusion Matrix (to visualize classification results)

b) ROC Curve (to evaluate model performance)

c) Feature Importance Graph

5. Required topics from the subject:

Arrays: Used to store and process dataset values

Files: Reading dataset (CSV files)

Functions: Modular coding for preprocessing and training

Basic Data Structures: Handling structured data efficiently

6.Platform Required:

- **Jupyter Notebook / Google Colab / Python IDLE**

7.Books Sources:

a) *Hands-On Machine Learning with Scikit-Learn* – Aurélien Géron

b) *Python for Data Analysis* – Wes McKinney

c) Scikit-learn official documentation

d) Kaggle (dataset and tutorials)

e) GeeksforGeeks (ML concepts)